

Christophe de Dinechin

Software as a Passion

SYSTEMS | LANGUAGES | TOOLS | GAMES | RESEARCH | COMMUNICATION SKILLS

ABOUT

Pushing the boundaries by writing highly innovative and differentiating software that makes a real difference.

French software developer — operating systems, languages, games

Amateur physics researcher and tinkerer in strange forms of art

KEY SKILLS

- Systems & Virtualization
- Languages & Compilers
- Tools & Apps
- Games & 3D
- Research
- Communication Skills

TECHNOLOGIES

C, C++, Virtualization, Kernel, Real-time, 3D, AI, SoC, Validation, Performance, Linux, macOS, OpenShift, RPL, XL, Tao3D, ELFE, SPICE, Make

INFO

- MS Computer Science, École des Mines (1991, Paris)
- christophe@dinechin.org
- github.com/c3d
- c3d.github.io

Scan the constellation
https://c3d.github.io/cv/

SYSTEMS & VIRTUALIZATION

HPVM - Enterprise-grade Itanium virtualization 2000-2010
- From first line of code and pre-silicon bring up all the way to 7 digits of revenue per quarter

HP ECUTEST Real-Time System 1994-1998
- Designed HP ECUTEST for engine control unit testing and system validation

SPICE Remote Streaming 2017-Present
- Part of the Spice team at Red Hat

DxO ONE Connected Camera 2015-2016
- Dual-core ARM + ISP real-time system

LANGUAGES & COMPILERS

HP aC++ Compiler 1998-2000
- Wrote the first native Itanium C++ compiler, pre-silicon testing on ski, overriding HP's plans to ship a PA-RISC to Itanium cross-compiler running in the Aries PA-RISC to Itanium dynamic translator
- Exception handling (reference implementation, WIESS paper), C++ ABI initiator, C++ standards committee
- Performance (symbol table, precompiled headers) and build system harmonised across 16 teams

DB48X - Modern RPL Calculator 2022-Present
- Best scientific calculator today, a modern rebuild of high-end HP-48 and HP-50 calculators

Tao3D 2010-Present
- Programming language for interactive 3D, derived from XLR

XL 1992-Present
- Renamed from LX after discussions notably with Daveed Vandevoorde; focus shifted from experimental to extensible

TOOLS & APPS

Make-It-Quick
- Short readable makefiles without automake/cmake complexity

Recorder
- Continuous recording with printf-like statements for performance and debug traces

Emacs on Aqua
- Adapted an old NeXT port to Aqua when only Terminal Emacs shipped

GAMES & 3D

Alpha Waves / Continuum 1989-1990
- Started in 1988, published by Infogrames in 1990

HP-48 Games 1989-1993
- PacMan — first use of HP48 hardware scroll (per Paul Courbis)

Taodyne 2010-2016
- Created Tao3D to present in real-time 3D (big data, movies, glasses-free 3D)

RESEARCH

Computer science research
- Published CS research: WIESS 2000 C++ exception handling; FOSDEM 2024 / FOSDEM 2023 talks on DB48X

12 patents awarded, 3 pending
- 12 granted US patents (HP and Red Hat, virtualization and confidential computing), 3 pending applications including split host/tenant cluster credentials — Google Patents (de Dinechin / Dupont de Dinechin)

Physics research
- A Theory of Incomplete Measurements — unifying GR and QM (TIM, Réunifions la physique)